

SHIHAB BIN KADER

Computer Vision | Multimodal Learning | Explainable & Reliable AI

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RESEARCH PROFILE

Computer Science and Engineering graduate with research and development experience in computer vision, deep learning, multimodal AI, and explainable AI. Co-conducted a 6,000-image classification study using CNNs, transfer learning, and Grad-CAM, and built reproducible ML applications with PyTorch, TensorFlow/Keras, OpenCV, Hugging Face, Streamlit, and Git. Interested in multimodal representation learning across vision and language, foundation models, uncertainty-aware learning, and AI for biodiversity/science.

EDUCATION

BSc in Computer Science and Engineering, East Delta University, Chattogram, Bangladesh Sep 2020 - Jan 2025
CGPA: 3.13/4.00 | 162 credits | Joint thesis: *Mango Ripeness Detection Using Deep Learning with XAI (Grad-CAM)*
Relevant coursework: Pattern Recognition, Digital Image Processing, Machine Learning, Artificial Intelligence, Probability & Statistics, Big Data Analytics

RESEARCH EXPERIENCE

Joint Undergraduate Thesis Researcher | East Delta University 2024 - 2025

- Co-conducted six-stage mango-ripeness classification using a balanced 6,000-image dataset, preprocessing/augmentation, an 80/10/10 split, CNNs, and transfer-learning architectures.
- Evaluated accuracy, precision, recall, F1, confusion matrices, ROC/AUC, class-wise errors, and confidence behaviour; used Grad-CAM to inspect visual evidence and model attention.
- Worked across dataset curation, experiment design, visual error analysis, interpretation, and research writing.

SELECTED PUBLICATION

Abdullah, M. S., Faiyaz, G. M. F., Forhad, O. N., **Kader, S. B.**, & Labib, S. M. A. (2025). "Hybrid Xception-LSTM Model for Explainable Mango Ripeness Classification Using Grad-CAM." IEEE 4th RAAICON, Dhaka, Bangladesh. [Fourth author]
Contribution: literature review, data analysis, result interpretation, and manuscript review/revision.

SELECTED AI / VISION PROJECTS

ModelDNA - Explainable AI Toolkit | Python, PyTorch, Streamlit, OpenCV, Grad-CAM

- Built an interactive toolkit for class-activation heatmaps, confidence analysis, and comparison of model-attention and failure patterns.

Multimodal AI Application | Python, Streamlit, Hugging Face Transformers

- Integrated image captioning, PDF question answering, and chatbot workflows in one interface for heterogeneous visual and textual inputs.

Image Caption Generator | Python, InceptionV3, LSTM

- Implemented a visual-language pipeline combining pretrained image features with sequence generation and qualitative error analysis.

Sign Sense - Real-Time ASL Detector | Python, TensorFlow/Keras, OpenCV

- Built a 29-class webcam recognition application and analysed confusion among visually similar gestures in a real-time inference pipeline.

PROFESSIONAL EXPERIENCE

Information Management Officer | ACTED, Cox's Bazar, Bangladesh Jul 2025 - Jun 2026

- Cleaned, validated, and analysed operational datasets; produced dashboards, GIS maps, and analytical reports while coordinating multi-partner data consolidation and quality checks.

TECHNICAL SKILLS

ML/ Vision: PyTorch, TensorFlow/Keras, OpenCV, CNNs, LSTMs, transfer learning, Grad-CAM, feature extraction, model evaluation | **Programming / Data:** Python, Pandas, NumPy, SQL, R, Java, C/C++

Research/Development: Hugging Face Transformers, Jupyter, Matplotlib, Streamlit, FastAPI, Git/GitHub, Docker; dataset curation, experimental splits, error analysis, reproducible reporting

HONOURS & ENGLISH PROFICIENCY

Dean's List Recognition, East Delta University | EDU Academic Scholarship | IELTS Academic 7.5 (L8.0, R8.0, W6.5, S7.0), CEFR C1, Jan 2026